

ASTORIA, OR, USA

WMO: 727910

Lat: 46.157N

Lon: 123.883W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.9	-0.5	-8.5	1.8	1.2	-5.1	2.5	2.1	14.3	11.6	12.5	11.3	2.5	80	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	25.0	17.5	22.4	16.7	20.7	16.2	18.4	23.5	17.5	21.3	16.7	19.9	4.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	19.6	15.8	11.2	18.9	15.2	10.8	18.2	51.9	23.3	48.9	21.3	46.7	19.8	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	8.9	7.9	-4.6	31.3	2.0	2.3	-6.0	33.0	-7.2	34.3	-8.4	35.6	-9.8	37.2		
(5)			-5.6	21.1	2.2	1.0	-7.1	21.8	-8.4	22.4	-9.6	22.9	-11.2	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	6.8	6.8	7.9	9.4	11.9	14.1	16.0	16.4	15.1	11.7	8.7	6.4	(6)
(7)		DBStd	4.34	2.87	2.70	2.36	2.30	2.31	1.92	1.61	1.59	2.17	2.52	3.04	3.03	(7)
(8)		HDD10.0	498	104	92	73	37	6	0	0	0	0	11	59	116	(8)
(9)		HDD18.3	2706	358	322	323	269	201	129	75	64	101	204	290	370	(9)
(10)		CDD10.0	850	4	3	8	18	66	123	187	198	153	66	19	4	(10)
(11)		CDD18.3	13	0	0	0	0	1	2	3	4	3	0	0	0	(11)
(12)		CDH23.3	166	0	0	0	5	18	24	32	44	40	3	0	0	(12)
(13)		CDH26.7	42	0	0	0	1	5	7	9	12	8	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.5	3.5	3.6	3.4	3.3	3.4	3.5	3.1	2.9	3.0	3.4	3.7	(14)
(15)	Precipitation	PrecAvg	1737	264	188	188	133	83	61	26	35	68	154	271	271	(15)
(16)		PrecMax	2355	612	556	389	256	183	139	112	137	185	342	498	518	(16)
(17)		PrecMin	1056	18	34	26	42	9	14	0	1	1	13	37	68	(17)
(18)		PrecStd	310	119	99	73	52	36	32	25	31	48	82	99	99	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	15.7	18.8	22.7	25.6	26.5	27.5	28.1	27.6	22.9	17.6	14.9	(19)
(20)			MCWB	11.5	9.8	11.6	14.3	16.5	18.5	19.5	19.4	17.7	15.2	13.8	12.9	(20)
(21)		2%	DB	13.0	13.6	15.1	17.5	20.4	21.5	22.7	23.5	23.8	19.3	15.5	13.0	(21)
(22)			MCWB	10.7	9.6	10.5	12.3	14.7	16.4	17.5	17.9	17.1	14.0	13.1	11.5	(22)
(23)		5%	DB	12.0	12.0	13.1	14.8	17.4	19.1	20.8	21.3	21.1	17.3	14.4	11.8	(23)
(24)			MCWB	10.4	9.6	10.1	11.0	13.7	15.0	16.6	16.9	16.1	14.0	12.3	10.4	(24)
(25)		10%	DB	11.0	10.9	11.8	13.1	15.7	17.6	19.5	20.0	19.2	15.9	13.4	10.9	(25)
(26)			MCWB	9.6	9.1	9.4	10.2	12.5	14.1	15.8	16.2	15.4	13.4	11.6	9.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.3	12.7	14.9	17.4	18.8	19.8	19.8	19.2	16.9	14.9	13.2	(27)
(28)			MCDWB	13.8	13.5	15.7	21.4	23.8	25.9	26.1	27.0	24.9	19.6	16.4	14.1	(28)
(29)		2%	WB	11.7	11.1	11.6	12.8	15.2	16.7	17.9	18.2	17.7	15.5	13.7	11.9	(29)
(30)			MCDWB	12.3	12.2	13.7	16.1	19.4	21.2	22.2	22.9	21.6	17.7	14.9	12.7	(30)
(31)		5%	WB	10.7	10.2	10.7	11.6	13.9	15.5	16.9	17.4	16.8	14.5	12.9	10.8	(31)
(32)			MCDWB	11.5	11.4	12.5	14.0	17.0	18.4	20.1	20.8	20.0	16.4	14.0	11.5	(32)
(33)		10%	WB	9.9	9.1	10.0	10.7	12.8	14.6	16.2	16.7	16.0	13.7	11.9	9.8	(33)
(34)			MCDWB	10.8	10.5	11.6	12.6	15.1	17.1	18.9	19.5	18.6	15.5	12.9	10.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.8	7.2	7.4	7.2	6.9	7.0	7.6	8.8	8.0	6.7	5.6	(35)
(36)			MCDBR	6.4	8.7	10.0	11.1	11.7	10.5	10.1	11.1	13.2	10.7	7.3	5.9	(36)
(37)			MCWBR	4.9	5.7	6.3	6.7	7.1	6.1	5.9	6.5	7.4	6.5	5.1	5.0	(37)
(38)		5% WB	MCDWB	5.4	6.4	8.0	9.2	10.0	9.5	9.2	9.9	10.6	8.0	5.4	5.3	(38)
(39)			MCWBR	4.8	4.9	5.6	6.0	6.3	5.8	5.5	6.0	6.3	5.5	4.4	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.319	0.336	0.367	0.387	0.367	0.352	0.357	0.346	0.336	0.327	0.322	(40)	
(41)		taud	2.485	2.554	2.519	2.394	2.358	2.440	2.510	2.493	2.530	2.550	2.523	2.495	(41)	
(42)		Ebn at Noon	769	866	897	891	880	898	908	891	873	831	768	737	(42)	
(43)		Edh at Noon	68	76	90	111	119	111	102	100	88	75	65	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.12	1.98	2.96	4.29	5.30	5.79	6.20	5.33	4.14	2.45	1.33	0.94	(44)	
(45)		RadStd	0.17	0.25	0.35	0.38	0.47	0.49	0.49	0.42	0.35	0.28	0.17	0.13	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+25

Nomenclature: See separate page